

# NAMAN MANOCHA

AI ENGINEER • GENAI • AGENTIC SYSTEMS — LLMs, RAG & Applied ML

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## SUMMARY

AI/ML engineer with hands-on experience across classical ML, multi-agent LLM orchestration, and grounded RAG systems, with a track record of shipping production AI products end-to-end. Solo-built **FundersAI** (live stock & mutual fund research platform with 7-node LangGraph RAG, explainable clustering, and a decoupled report-synthesis microservice on K3s/AWS), **TalentOS** (autonomous opportunity intelligence agent on Google ADK/LangGraph with 91% pre-filtering), and **CareFlow** (clinical RAFT fine-tuning with 88.5% precision). Experienced translating business requirements into deployed systems as a technology lead; comfortable across Python, FastAPI, Next.js, Docker, Kubernetes, and pgvector.

## TECHNICAL SKILLS

**ML & AI:** LLMs, RAG (LangGraph), Multi-Agent Orchestration (LangGraph, Google ADK), Hybrid Search, Vector Embeddings (pgvector, ChromaDB), Fine-Tuning (QLoRA, Unsloth), Prompt Engineering, scikit-learn, MLflow, Langfuse

**Backend & Infra:** Python, FastAPI, REST APIs, SQL, PostgreSQL, Supabase (Auth, RLS), Redis, Docker, Kubernetes (K3s), AWS EC2, Google Cloud Run, GitHub Actions (CI/CD), Git

**Frontend & Data:** Next.js, React, TypeScript, ETL Pipelines, Web Scraping, PDF Parsing, Data Validation, Data Visualization

## EXPERIENCE

### Technology Manager — PayGain Multiservices Pvt. Ltd. Feb 2025 – Present

Kolkata, India

Scoping and leading development of a company-wide mobile app as sole engineer, covering client/employee onboarding, document access, product registration, and payments — mobile counterpart to PayGain core platform.

Designed and deployed lead-capture automation (Make); drove 30+ client conversions via website and digital discovery flows; simplified complex financial products for 300+ seminar attendees via 20+ technical presentations.

## INDEPENDENT PROJECTS

### FundersAI — AI-Powered Research Platform for Indian Stocks & Mutual Funds

[fundersai.co.in](https://fundersai.co.in)

*Python, FastAPI, Next.js, Supabase/pgvector, LangGraph, Groq, OpenAI, Cloudflare R2, Kubernetes (K3s)*

*Solo-built, Apr–Aug 2026 — ~143K LOC across 864 files, 431 commits, 167+ test suites — OpenAI Build Week | Live:*

*[www.fundersai.co.in](https://www.fundersai.co.in) | Repo: [github.com/Naman6019/FundersAI](https://github.com/Naman6019/FundersAI)*

Citation-Grounded RAG & Multi-Agent Routing: built 7-node LangGraph pipeline (retrieve → grade → synthesize/abstain with rewrite loop) over ~10,400 chunks with lexical reranking and strict abstention; designed Groq LLM router for stock/fund/news paths.

Explainable ML & Microservice Architecture: engineered fund-similarity clustering (k-means, 13 features) and logistic regression triage; extracted report synthesis into a decoupled FastAPI/LangGraph microservice deployed on a 2-replica K3s cluster on AWS EC2.

Resilient Data Engineering: built 22 GitHub Actions workflows scraping/normalizing disclosures from 12 AMCs (99.66% data readiness across 871 funds); fixed normalizer bug restoring coverage to 100% across 14,000+ records.

Cloud Storage & Billing: built an S3-compatible storage layer (Cloudflare R2, boto3/AWS SigV4) for raw documents and cold archival; shipped Razorpay subscription billing (webhook-verified, Free/Pro/Ultra tiers) with Supabase RLS enforcement.

### TalentOS — Autonomous Opportunity Intelligence Platform (All Things Agentic Hackathon)

*Python, Google ADK, LangGraph, Next.js, Firestore, Google Cloud Run | ~23K LOC across 131 files, 70 commits, 235+*

*test suite | Live: [all-things-agentic--allthingsage...](https://all-things-agentic--allthingsage...) | Repo: [github.com/Naman6019/all-things-agentic](https://github.com/Naman6019/all-things-agentic)*

Dual-Pipeline Multi-Agent System: built Careers and Studio pipelines (LangGraph StateGraphs orchestrating Google ADK agents on Gemini) ingesting ~2,500 postings/run across 8 ATS sources; applies deterministic pre-filtering cutting 91% before LLM calls.

Guardrails & Full-Stack Delivery: shipped human-in-the-loop send gates, Langfuse tracing, 235+ test hermetic pytest suite, and Next.js 15 dashboard with tailored resume generation deployed on Google Cloud Run.

### CareFlow Intelligence — Agentic RAG & Fine-Tuning for Clinical Research

*Python, FastAPI, Next.js, Supabase/pgvector, ChromaDB, Unsloth, Ollama | ~7,300 LOC — synthetic data (MITRE Synthea) | Repo:*

*[github.com/Naman6019/CareFlow-Intelligence](https://github.com/Naman6019/CareFlow-Intelligence)*

RAFT Fine-Tuning & Local Serving: designed RAFT pipeline with oracle evidence and hard distractors; fine-tuned Qwen2.5-7B via 4-bit QLoRA (Unsloth) achieving 88.5% citation precision and 84.6% abstention; served locally via quantized GGUF on Ollama.

Agentic RAG & Ingestion: built bounded-tool citation-grounded RAG agent over 4 ingestion pipelines with Postgres GIN citation checks, plus a human-in-the-loop ingestion agent managing 18 FHIR R4 tables.

Human-in-the-Loop Data Agent: built a second agent for synthetic clinical data ingestion (10 patients, 479 encounters, 5,368 observations across 18 FHIR R4 tables) with mandatory human-approval gates; added an OpenRouter (Nemotron 3 Ultra) cloud fallback for the RAG agent.

### Other Projects & Research

SQuAD QA Benchmarking & NLP Research: compared BERT, BiDAF, and DistilBERT for extractive QA; fine-tuned each with standardized eval pipeline (precision, recall, F1); published research paper in IJMDES.

GitResume Agent: self-built multi-agent LLM pipeline (PyPI: git-resume-agent, ~4K LOC, 26 commits) that synthesizes and adversarially verifies grounded resume bullets from git history.

## EDUCATION & ACHIEVEMENTS

**B.Tech, Computer Science Engineering (AI & ML Specialization) — VIT Amaravati**

**June 2021 – Sept 2025**

CGPA: 8.2/10 | Published Research Paper in IJMDES | Certifications: Azure Administrator Associate, AWS Cloud Practitioner